

Nombre:

Día

Mes

Año

Folio

Tema:

Cal. 10

Tarea 20

1. Calcular los siguientes límites

$$a) \lim_{x \rightarrow \frac{\pi}{4}} \sin 2x = \sin 2\left(\frac{\pi}{4}\right) = \sin\left(\frac{\pi}{2}\right) = 1$$

$$b) \lim_{x \rightarrow 0} \frac{\sin 2x - 3 \cos x}{1 + x} = \frac{\sin 2(0) - 3 \cos(0)}{1 + 0} = \frac{\sin(0) - 3 \cos(0)}{1} = -3(1) = -3$$

$$c) \lim_{x \rightarrow \frac{\pi}{2}} \frac{\operatorname{tg} \frac{x}{2} - \cos 2x}{\sin x + 1} = \frac{\operatorname{tg}\left(\frac{\pi/2}{2}\right) - \cos 2\left(\frac{\pi}{2}\right)}{\sin\left(\frac{\pi}{2}\right) + 1} = \frac{\operatorname{tg}\left(\frac{\pi}{4}\right) - \cos \pi}{\sin\left(\frac{\pi}{2}\right) + 1} = \frac{1 - (-1)}{1 + 1} = \frac{1 + 1}{2} = \frac{2}{2} = 1$$

$$d) \lim_{w \rightarrow 0} \frac{\operatorname{tg}^2 w - 1}{\operatorname{tg}^2 w + 1} = \frac{\operatorname{tg}^2(0) - 1}{\operatorname{tg}^2(0) + 1} = \frac{0 - 1}{0 + 1} = -1$$

$$e) \lim_{\beta \rightarrow \frac{\pi}{3}} \frac{\sin \beta - \cos \beta}{\operatorname{tg} \beta - \sqrt{3}} = \frac{\frac{\sqrt{3}}{2} - \frac{1}{2}}{\sqrt{3} - \sqrt{3}} = \frac{\frac{\sqrt{3} - 1}{2}}{0} = \frac{\sqrt{3} - 1}{0} \rightarrow \infty$$

$$\frac{\sin\left(\frac{\pi}{3}\right) - \cos\left(\frac{\pi}{3}\right)}{\operatorname{tg}\left(\frac{\pi}{3}\right) - \sqrt{3}}$$